

Presentation Script

Sign Language Recognition System — RVU SOCSE

SLIDE 9 — Key Innovations

Okay guys, simple thing.

Deaf people use hand signs to talk. Cool. But the other person needs to understand it also right? That's the problem.

So we said — camera sees the hand, computer reads it, screen shows the word. Done. **That's the whole project.**

Built with three ideas — dedicated hardware display, dirt cheap components, and expandable design. Simple, affordable, future-proof.

Total cost? **Under two thousand rupees.** Less than your last Swiggy order. Significantly more useful.

SLIDE 10 — Technical Pipeline

[CLICK after 'Swiggy order' line]

Now how does it actually work. Eight stages. Stay with me.

Webcam grabs a frame every thirty-three milliseconds. Gets converted, histogram equalized, Gaussian blurred — basically the image hits the gym before we use it.

Then MediaPipe CNN finds **twenty-one landmarks on your hand** — every joint, every fingertip, mapped precisely. Anything below fifty percent confidence? **Thrown out. No mercy.**

Those landmarks become **two hundred and nine features** — inter-landmark distances, joint angles, orientation values. All normalized, all clean.

That goes into KNN or Random Forest. Classified. Confidence below seventy-five percent? Rejected again.

Like Ranveer Singh in Dhurandhar — **ghayal hoon isliye ghatak hoon**. Our system is wounded by bad data and deadly accurate because of it.

[PAUSE — let them laugh]

Result fires over UART at **one lakh fifteen thousand two hundred baud** to ESP32. Renders on the display at ten hertz via SPI.

Total latency — fifty to hundred milliseconds. Real time.

SLIDE 11 — Expected Outcomes

[CLICK after 'Real time']

What do we get?

Thirty plus frames per second. Over ninety percent accuracy. Hardware and software talking properly. Fully documented prototype — **portable, standalone, under two thousand rupees.**

Hearing-impaired people deserve good technology without spending a fortune.

Camera. Chip. Classified. Job done.

Thank you. Don't ask for notes.

Slide Cue Summary

Slide 9 — Opening — after 'That's the whole project'

Slide 10 — After 'Less than your last Swiggy order'

Slide 11 — After 'Real time'

~ 75 seconds total. Pause after the Dhurandhar line. Everything else — keep moving.